



Translated

From: e-maxx.ru

## Factorial modulo $p$

In some cases it is necessary to consider complex formulas modulo some prime  $p$ , containing factorials in both numerator and denominator, like such that you encounter in the formula for Binomial coefficients. We consider the case when  $p$  is relatively small. This problem makes only sense when the factorials appear in both numerator and denominator of fractions. Otherwise  $p!$  and subsequent terms will reduce to zero. But in fractions the factors of  $p$  can cancel, and the resulting expression will be non-zero modulo  $p$ .

Thus, formally the task is: You want to calculate  $n! \bmod p$ , without taking all the multiple factors of  $p$  into account that appear in the factorial. Imagine you write down the prime factorization of  $n!$ , remove all factors  $p$ , and compute the product modulo  $p$ . We will denote this *modified factorial* with  $n!_{(\%p)}$ . For instance  $7!_{(\%3)} \equiv 1 \cdot 2 \cdot \underbrace{3}_{\%3} \cdot 4 \cdot 5 \cdot \underbrace{6}_{\%3} \cdot 7 \equiv 2 \bmod 3$ .

Learning how to effectively calculate this modified factorial allows us to quickly calculate the value of the various combinatorial formulas (for example, [Binomial coefficients](#)).

## Algorithm

Let's write this modified factorial explicitly.

$$\begin{array}{l} n!_{\substack{\text{mod } p}} = 1 \cdot 2 \cdot 3 \cdot \dots \cdot (p-2) \cdot (p-1) \cdot \\ \underbrace{1}_{\substack{\text{mod } p}} \cdot (p+1) \cdot (p+2) \cdot \dots \cdot (2p-1) \cdot \underbrace{2}_{\substack{\text{mod } p}} \cdot \\ \dots \cdot (2p+1) \cdot \dots \cdot (p^2-1) \cdot \underbrace{1}_{\substack{\text{mod } p}} \cdot (p^2+1) \\ \cdot \dots \cdot n \pmod{p} \end{array} = 1 \cdot 2 \cdot 3 \cdot \dots \cdot (p-2) \cdot (p-1) \cdot \underbrace{1}_{\substack{\text{mod } p}} \cdot 1 \cdot 2 \cdot \dots \cdot (p-1) \cdot \underbrace{2}_{\substack{\text{mod } p}} \cdot 1 \cdot 2 \cdot \dots \cdot (p-1) \cdot \underbrace{1}_{\substack{\text{mod } p}} \cdot (p^2) \cdot 1 \cdot 2 \cdot \dots \cdot (n \bmod p) \pmod{p}$$

It can be clearly seen that factorial is divided into several blocks of same length except for the last one.

$$n!_{\text{\scriptsize (p)}} = \underbrace{1 \cdot 2 \cdot 3 \cdot \ldots \cdot (p-2) \cdot (p-1)}_{\text{\scriptsize (1)}} \cdot \underbrace{1 \cdot 2 \cdot 3 \cdot \ldots \cdot (p-2) \cdot (p-1)}_{\text{\scriptsize (2)}} \cdot \ldots \cdot \underbrace{1 \cdot 2 \cdot 3 \cdot \ldots \cdot (p-2) \cdot (p-1)}_{\text{\scriptsize (p-1)}}$$